

Tensor decompositions for data analysis

Module 2 of 4

Applied Math 210: Algebraic Fundamentals of Representing Data

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Introduction

The first part of the course studied factorizations of a covariance matrix or data matrix. The former arises when modeling an observed random vector $\mathbf{x} \in \mathbb{R}^p$ as a linear transformation of a latent random vector, $\mathbf{x} = A\mathbf{z}$, and recovered A from the covariance $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^T$ under the assumption that the entries of $\mathbf{z}$ are *uncorrelated*. This is a matrix factorization problem, solved by the spectral theorem, and it identifies A up to an orthogonal transformation: any $A' = AQ$, for Q orthogonal, gives a model with equally good fit. This module of the course replaces “uncorrelated” with something stronger, the assumption that the entries of $\mathbf{z}$ are independent. This looks beyond the covariance matrix, at higher-order statistics. The higher-order statistics are *tensors*: multi-indexed arrays that generalize matrices. A random vector’s covariance is its second cumulant; its third, fourth, and so on, cumulants are tensors, and diagonalizing these gives new factorizations for data analysis. Tensors also show up as arrays of data that are organized via multiple indices, such as a tensor of neurons by experiment by time in neuroscience.

This module builds the algebra of tensors: what they are, what their rank means, when a tensor is generic, and how existence and uniqueness of a tensor decomposition are proved. It then studies independent component analysis (ICA), culminating in Comon’s identifiability theorem: the ICA model is identifiable if and only if at most one source is Gaussian. The same three questions from Module 1 (expressivity, identifiability, optimization landscapes) reappear, with a new answer to the second: for tensors of order at least three, a generic low-rank decomposition is unique, with no extra structural assumption required (such as orthogonality) unlike the matrix case.

Several topics are beyond the scope of this module, including tensor decomposition algorithms in the numerical-linear-algebra sense (such as alternating least squares, Jennrich’s algorithm, or tensor power iteration), and the use of moment- and cumulant-tensor methods to estimate latent-variable and mixture models beyond ICA. Both are descendants of the ideas developed here.

Useful texts

This module studies the algebra of tensors, motivated by questions about identifying a linear latent-variable model from higher-order statistics or modules from tensors of data. J. M. Landsberg’s *Tensors: Geometry and Applications* develops tensor rank, symmetric rank, and the algebraic geometry of secant varieties. Kolda and Bader’s survey *Tensor Decompositions and Applications* and Hyvärinen, Karhunen, and Oja’s book *Independent Component Analysis* take a more applied view, stating the results we prove here (Kruskal’s uniqueness theorem; Comon’s identifiability theorem). Lim’s *Tensors in computations* (Acta Numerica, 2021) organizes the definition of a

tensor around three ideas (equivariance, multilinearity, separability) a useful third lens on the three definitions we open with. A related reference is my article “Factorizations for Data Analysis,” *SIAM News*, Vol. 58, No. 7, 2025, which covers ICA, briefly.

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1 Notation

We continue Module 1’s convention: bold lowercase for vectors (fixed or random), plain lowercase for scalars, plain uppercase for matrices. Tensors of any order are also set in plain uppercase. A subscript on a bold symbol labels a vector, and a superscript in parentheses labels which factor of a tensor (or vector space) is meant.

Symbol	Meaning
$\mathbf{x} = (x_1, \dots, x_p)^\top$	random vector of p observed variables
$\mathbf{z} = (z_1, \dots, z_q)^\top$	random vector of q latent (source) variables
$A \in \mathbb{R}^{p \times q}$	mixing matrix, $\mathbf{x} = A\mathbf{z}$; column $\mathbf{a}_k$ is source k 's per-variable loading
$T \in \mathbb{R}^{n_1} \otimes \dots \otimes \mathbb{R}^{n_d}$	a tensor of order d and format $n_1 \times \dots \times n_d$; entries $T_{i_1 \dots i_d}$
$S^d(\mathbb{R}^n)$	the space of symmetric order- d tensors on $\mathbb{R}^n$
$\mathbf{u}^{(1)} \otimes \dots \otimes \mathbf{u}^{(d)}$	a rank-one tensor; entries $u_{i_1}^{(1)} \dots u_{i_d}^{(d)}$
$\mathbf{v}^{\otimes d}$	the symmetric rank-one tensor $\mathbf{v} \otimes \dots \otimes \mathbf{v}$ (d times)
$\text{rank}(T)$	tensor rank: fewest rank-one summands
$\text{rank}_+(T)$	non-negative rank: fewest rank-one summands with non-negative factors
k_A	Kruskal rank of A : largest k such that every k columns of A are linearly independent
$\mathcal{M}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$	d -th moment tensor of $\mathbf{z}$, $\mathbb{E}[\mathbf{z}^{\otimes d}]$
$\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$	d -th cumulant tensor of $\mathbf{z}$, $\mathbb{E}[\mathbf{z}^{\otimes d}]$ up to lower order terms
$A \bullet S$	the action of $A \in \mathbb{R}^{p \times q}$ on $S \in S^d(\mathbb{R}^q)$, generalizing the congruence $A\Sigma A^\top$
$T(\mathbf{u}, \mathbf{v}, \mathbf{w})$	multilinear-map evaluation, $\sum_{ijk} T_{ijk} u_i v_j w_k$ (similarly for other orders)

2 What Is a Tensor?

To motivate the definition of a tensor, it helps to first ask the same of a matrix, since each answer has an order- d generalization. A matrix $M \in \mathbb{R}^{n_1 \times n_2}$ is:

- (i) a 2D array of numbers, $(M_{ij})_{1 \leq i \leq n_1, 1 \leq j \leq n_2}$;
- (ii) a linear map $\mathbb{R}^{n_2} \rightarrow \mathbb{R}^{n_1}$;
- (iii) an element of the tensor product space $\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}$, built from the outer product $\mathbf{u} \otimes \mathbf{v} := \mathbf{u}\mathbf{v}^\top$ of two vectors $\mathbf{u} \in \mathbb{R}^{n_1}$, $\mathbf{v} \in \mathbb{R}^{n_2}$, via $(\mathbf{u} \otimes \mathbf{v})_{ij} = u_i v_j$.

A tensor of order d is the same object with d indices instead of 2: a d -way array; a multilinear map of $d - 1$ vector arguments; an element of a d -fold tensor product space. We take each in turn.

Definition 2.1 (Tensor, order, format). A *tensor of order d* is an array T with entries $T_{i_1 \dots i_d}$, $i_1 \in \{1, \dots, n_1\}, \dots, i_d \in \{1, \dots, n_d\}$. We say T has *format* $n_1 \times \dots \times n_d$, and write $T \in \mathbb{R}^{n_1} \otimes \dots \otimes \mathbb{R}^{n_d}$.

Order $d = 1$ is a vector; order $d = 2$ is a matrix.

Definition 2.2 (Outer product, tensor product space). For vectors $\mathbf{u}^{(1)} \in \mathbb{R}^{n_1}, \dots, \mathbf{u}^{(d)} \in \mathbb{R}^{n_d}$, their *outer product* $\mathbf{u}^{(1)} \otimes \dots \otimes \mathbf{u}^{(d)}$ is the order- d tensor with entries

$$(\mathbf{u}^{(1)} \otimes \dots \otimes \mathbf{u}^{(d)})_{i_1 \dots i_d} = u_{i_1}^{(1)} \dots u_{i_d}^{(d)}.$$

The outer product is multilinear: $(\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2) \otimes \mathbf{v} = \lambda_1 (\mathbf{u}_1 \otimes \mathbf{v}) + \lambda_2 (\mathbf{u}_2 \otimes \mathbf{v})$, and likewise in each other argument. The *tensor product space* $\mathbb{R}^{n_1} \otimes \dots \otimes \mathbb{R}^{n_d}$ is the space of all finite sums $\sum_k \mathbf{u}_k^{(1)} \otimes \dots \otimes \mathbf{u}_k^{(d)}$.

A tensor is also a multilinear map: T of format $n_1 \times \dots \times n_d$ defines $T(\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(d)}) := \sum_{i_1 \dots i_d} T_{i_1 \dots i_d} u_{i_1}^{(1)} \dots u_{i_d}^{(d)} = \langle T, \mathbf{u}^{(1)} \otimes \dots \otimes \mathbf{u}^{(d)} \rangle$, linear in each argument (where here $\langle A, B \rangle = \sum A_{i_1 \dots i_d} B_{i_1 \dots i_d}$ is the entrywise inner product). Equivalently, T can be viewed as a linear map $\mathbb{R}^{n_1} \otimes \dots \otimes \mathbb{R}^{n_{d-1}} \rightarrow \mathbb{R}^{n_d}$, with the roles of any subset of indices grouped together in this way. The matrix case ($d = 2$) is the special case with one index on each side.

Definition 2.3 (Tensor rank). A non-zero tensor T has *rank one* if $T = \mathbf{u}^{(1)} \otimes \cdots \otimes \mathbf{u}^{(d)}$ for some vectors $\mathbf{u}^{(i)}$. It has *rank r* if $T = \sum_{k=1}^r \mathbf{u}_k^{(1)} \otimes \cdots \otimes \mathbf{u}_k^{(d)}$ for some rank-one tensors, and r is the fewest such number of terms.

Definition 2.4 (Symmetric tensor, symmetric rank). A tensor $T \in (\mathbb{R}^n)^{\otimes d}$ (i.e. $n_1 = \cdots = n_d = n$) is *symmetric* if $T_{i_1 \dots i_d} = T_{\sigma(i_1) \dots \sigma(i_d)}$ for every permutation σ of $\{1, \dots, d\}$. We write $S^d(\mathbb{R}^n)$ for the space of such tensors. A rank-one tensor is symmetric iff it has the form $\mathbf{v}^{\otimes d} := \mathbf{v} \otimes \cdots \otimes \mathbf{v}$ for some $\mathbf{v} \in \mathbb{R}^n$. A symmetric tensor S has *symmetric rank r* if $S = \sum_{k=1}^r \lambda_k \mathbf{v}_k^{\otimes d}$ for scalars λ_k and unit vectors $\mathbf{v}_k$, and r is fewest such.

For $d = 2$, symmetric rank is the notion of rank used there for symmetric matrices: for symmetric matrices, symmetric rank equals ordinary matrix rank (see Section 4, Remark 4.5). This is known to fail for real tensors, but remains open over the complex numbers.

3 Examples of Tensors

Tensor rank can hold different interpretations, depending on the context. We work through four examples.

Polynomials

Proposition 3.1 (Symmetric tensors are homogeneous polynomials). *There is a linear bijection between $S^d(\mathbb{C}^n)$ and homogeneous degree- d polynomials in n variables, sending $S \mapsto f_S$,*

$$f_S(\mathbf{x}) = \sum_{i_1, \dots, i_d} S_{i_1 \dots i_d} x_{i_1} \cdots x_{i_d}, \quad \mathbf{x} = (x_1, \dots, x_n)^\top.$$

We work over $\mathbb{C}$ in this section, since the genericity statements below (Alexander–Hirschowitz, Section 4) are true there. The map is well defined and linear by construction; it is a bijection because every homogeneous degree- d polynomial has a unique expansion in the monomial basis, and symmetry of S is the requirement needed to recover the entries of S from those coefficients (they are equal up to multinomial coefficients). If T is not symmetric, $T \mapsto f_T$ is still well defined but no longer injective: any tensor with the same symmetrization gives the same polynomial.

The polynomial f corresponding to the rank one tensor $\lambda \mathbf{v}^{\otimes d}$ is $\lambda \langle \mathbf{v}, \mathbf{x} \rangle^d$, a d -th power of a linear form. Extending by linearity:

Corollary 3.2. *A symmetric tensor $S \in S^d(\mathbb{C}^n)$ has symmetric rank $\leq r$ if and only if f_S is a sum of r d -th powers of linear forms (a Waring decomposition of f_S).*

A homogeneous polynomial of degree d in $n+1$ variables $x_1, \dots, x_n, x_{n+1}$ defines, via $x_{n+1} = 1$, a polynomial of degree $\leq d$ in n variables; its zero set is a hypersurface in $\mathbb{R}^n$ (or $\mathbb{C}^n$). A degree-3 hypersurface is a *cubic surface*.

Theorem 3.3 (Sylvester’s Pentahedral Theorem, 1851). *A general cubic surface f (equivalently, a general $T \in S^3(\mathbb{C}^4)$) can be written as*

$$f = \ell_1^3 + \cdots + \ell_5^3$$

for linear forms $\ell_1, \dots, \ell_5$ with complex coefficients.

This is consistent with the generic-rank formula we’ll get to later in Section 4: $\dim S^3(\mathbb{C}^4) = \binom{6}{3} = 20$, and $\lceil 20/4 \rceil = 5$.

Probability Mass Functions

Let $X_1, \dots, X_d$ be discrete random variables, with X_k taking m_k values, and let P be the joint probability mass function, viewed as a tensor of format $m_1 \times \dots \times m_d$: $P_{i_1 \dots i_d} = \Pr(X_1 = i_1, \dots, X_d = i_d)$.

Proposition 3.4. *P has rank 1 if and only if $X_1, \dots, X_d$ are independent.*

Proof. We give the argument for $d = 2$ for notational ease. If X_1, X_2 independent, $P_{ij} = \Pr(X_1 = i) \Pr(X_2 = j) = \mathbf{u} \mathbf{v}^\top$ for $u_i = \Pr(X_1 = i)$, $v_j = \Pr(X_2 = j)$: rank 1.

Conversely, suppose $P = \mathbf{u} \mathbf{v}^\top$ for some (not necessarily non-negative) $\mathbf{u}, \mathbf{v}$. Since $P_{ij} = u_i v_j \geq 0$ for all i, j , fixing any i_0 with $u_{i_0} \neq 0$ forces every v_j to share the sign of $u_{i_0}^{-1}$; then every u_i must in turn share a matching sign for $u_i v_j \geq 0$ to hold for all j . So $u_i v_j = |u_i| |v_j|$ for all i, j : $P = \mathbf{u}' (\mathbf{v}')^\top$ with $\mathbf{u}' = |\mathbf{u}|$, $\mathbf{v}' = |\mathbf{v}|$ entrywise non-negative. Since $\sum_{ij} P_{ij} = 1$, $\|\mathbf{u}'\|_1 \|\mathbf{v}'\|_1 = 1$. Setting $\mathbf{u}'' = \mathbf{u}' / \|\mathbf{u}'\|_1$, $\mathbf{v}'' = \mathbf{v}' / \|\mathbf{v}'\|_1$ gives $\mathbf{u}'' (\mathbf{v}'')^\top = \mathbf{u}' (\mathbf{v}')^\top / (\|\mathbf{u}'\|_1 \|\mathbf{v}'\|_1) = P$, and $u''_i = \sum_j P_{ij} = \Pr(X_1 = i)$, $v''_j = \Pr(X_2 = j)$: $P_{ij} = \Pr(X_1 = i) \Pr(X_2 = j)$, i.e. independence. $\square$

Non-negativity of the factors is required to extend the above to higher ranks.

Definition 3.5 (Non-negative rank). T has *non-negative rank* 1 if $T = \mathbf{u}^{(1)} \otimes \dots \otimes \mathbf{u}^{(d)}$ with every $\mathbf{u}^{(i)}$ entrywise non-negative. T has *non-negative rank* r if it is a sum of r non-negative-rank-1 tensors, and not fewer.

Proposition 3.6. *If $X_1, \dots, X_d$ are conditionally independent given an unobserved variable Z with r states, then P has non-negative rank $\leq r$. Consequently, for two variables, P has non-negative rank $\leq r$ if and only if X_1, X_2 are conditionally independent given some r -state Z .*

Proof. For $d = 2$: writing $\lambda_k = \Pr(Z = k)$, $a_{ik} = \Pr(X_1 = i \mid Z = k)$, $b_{jk} = \Pr(X_2 = j \mid Z = k)$,

$$P_{ij} = \sum_{k=1}^r \Pr(X_1 = i, X_2 = j \mid Z = k) \Pr(Z = k) = \sum_{k=1}^r \lambda_k a_{ik} b_{jk} = \left(\sum_{k=1}^r \lambda_k \mathbf{a}_k \otimes \mathbf{b}_k \right)_{ij},$$

a sum of r non-negative rank-one terms. Conversely, given $P = \sum_{k=1}^r \lambda_k \mathbf{a}_k \otimes \mathbf{b}_k$ with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$, and each $\mathbf{a}_k, \mathbf{b}_k$ a probability vector, introduce a latent Z with $\Pr(Z = k) = \lambda_k$, $\Pr(X_1 = i \mid Z = k) = a_{ik}$, $\Pr(X_2 = j \mid Z = k) = b_{jk}$. This yields conditional independence. $\square$

Exercise 3.7. Let $M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$. Show $\text{rank}(M) = 3$ (e.g. by exhibiting a linear dependence among the rows, and a non-vanishing 3×3 minor). It is a standard fact that $\text{rank}_+(M) = 4$: ordinary rank and non-negative rank differ.

Data Tensors

Many datasets are naturally indexed by three or more categories rather than two. These include: fluorescence spectroscopy (samples $\times$ emission wavelengths $\times$ excitation wavelengths), color images (height $\times$ width $\times$ RGB channel), or neural recordings (neuron $\times$ experiment $\times$ time). As with a data *matrix*, where rank counts the number of modules (Module 1, Section 2) combined in the data, the rank of a data *tensor* counts the number of underlying signals combined across all indexed contexts at once.

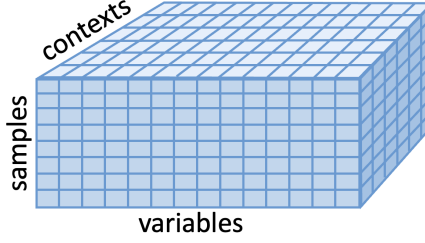


Figure 1: A data tensor, samples as rows, variables as columns, tubes as contexts

Matrix Multiplication

Definition 3.8 (Matrix multiplication tensor). For $A \in \mathbb{R}^{n_1 \times n_2}$, $B \in \mathbb{R}^{n_2 \times n_3}$, $C \in \mathbb{R}^{n_3 \times n_1}$, define the trilinear form

$$T_{\langle n_1, n_2, n_3 \rangle}(A, B, C) := \text{tr}(ABC).$$

Viewed as an element of $(\mathbb{R}^{n_1 \times n_2})^* \otimes (\mathbb{R}^{n_2 \times n_3})^* \otimes (\mathbb{R}^{n_3 \times n_1})^*$, this is the *matrix multiplication tensor*, of format $n_1 n_2 \times n_2 n_3 \times n_1 n_3$.

Proposition 3.9. $\text{rank}(T_{\langle n_1, n_2, n_3 \rangle}) \leq r$ if and only if there is an algorithm computing the entries of AB using r products of the form $\alpha_k(A)\beta_k(B)$, for linear forms α_k, β_k , followed only by additions.

Proof sketch. Suppose $T_{\langle n_1, n_2, n_3 \rangle} = \sum_{k=1}^r \alpha_k \otimes \beta_k \otimes \gamma_k$, so $\text{tr}(ABC) = \sum_{k=1}^r \alpha_k(A)\beta_k(B)\gamma_k(C)$ for all A, B, C . Fix A, B and set $\mu_k = \alpha_k(A)\beta_k(B)$ (computed with r products); then $\text{tr}(ABC) = \sum_k \mu_k \gamma_k(C)$ for every C . Taking C to be the elementary matrix E_{li} gives $\text{tr}(ABE_{li}) = (AB)_{il}$, so $(AB)_{il} = \sum_k \mu_k \gamma_k(E_{li})$: every entry of AB is a fixed (data-independent) linear combination of the r products μ_k . Conversely, an algorithm of this form determines such a rank- r decomposition. $\square$

The naive algorithm uses $r = n_1 n_2 n_3$ products (one per entry of A times one per entry of B , summed). For 2×2 matrices ($n_1 = n_2 = n_3 = 2$), this is $r = 8$.

Theorem 3.10 (Strassen, 1969). $\text{rank}(T_{\langle 2, 2, 2 \rangle}) \leq 7$.

Proof. Exhibit the 7 products directly: with $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$, set

$$\begin{aligned} M_1 &= (a_{11} + a_{22})(b_{11} + b_{22}), & M_2 &= (a_{21} + a_{22})b_{11}, & M_3 &= a_{11}(b_{12} - b_{22}), \\ M_4 &= a_{22}(b_{21} - b_{11}), & M_5 &= (a_{11} + a_{12})b_{22}, & M_6 &= (a_{21} - a_{11})(b_{11} + b_{12}), \\ M_7 &= (a_{12} - a_{22})(b_{21} + b_{22}). \end{aligned}$$

One checks that $(AB)_{11} = M_1 + M_4 - M_5 + M_7$, $(AB)_{12} = M_3 + M_5$, $(AB)_{21} = M_2 + M_4$, $(AB)_{22} = M_1 - M_2 + M_3 + M_6$; by Proposition 3.9, this is a rank-7 decomposition of $T_{\langle 2, 2, 2 \rangle}$. $\square$

It is an open problem to determine $\text{rank}(T_{\langle n_1, n_2, n_3 \rangle})$. In 2022, AlphaTensor (Fawzi et al., DeepMind), using reinforcement learning, found new low-rank decompositions of $T_{\langle n_1, n_2, n_3 \rangle}$ for several small formats. In 2025, AlphaEvolve (DeepMind) found a rank-48 decomposition of $T_{\langle 4, 4, 4 \rangle}$ for complex matrix multiplication, the first improvement since Strassen (1969) on the rank-49 bound obtained by recursively applying his own $T_{\langle 2, 2, 2 \rangle}$ decomposition.

The Matrix Multiplication Exponent

Definition 3.11. The *matrix multiplication exponent* ω is the infimum c such that $n \times n$ matrices can be multiplied using $O(n^c)$ arithmetic operations.

Recursively applying a rank- r decomposition of $T_{\langle n,n,n \rangle}$ in blocks gives an algorithm for $N \times N$ matrix multiplication using $O(N^{\log_n r})$ operations; this connects Proposition 3.9 to ω . The naive algorithm gives $\omega \leq 3$. Strassen's rank-7 decomposition of $T_{\langle 2,2,2 \rangle}$ gives $\omega \leq \log_2 7 \approx 2.807$. Successive refinements, including Coppersmith and Winograd (1990, $\omega < 2.376$), and a sequence of “laser method” improvements since, have brought the best known bound down to $\omega < 2.371339$ (Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou, 2025). Whether $\omega = 2$ (whether matrix multiplication is asymptotically as easy as reading the input) is a famous open problem.

4 The Generic Rank of a Tensor

Definition 4.1 (Generic). A property of points in $\mathbb{C}^N$ holds *generically* if it holds off a lower-dimensional (Zariski-closed) exceptional set. This implies that it holds with probability 1 under any continuous distribution on parameters.

For example: a generic $x \in \mathbb{R}$ is non-zero; for a fixed non-zero polynomial $f(x_1, \dots, x_n)$, a generic point $\mathbf{p} \in \mathbb{R}^n$ has $f(\mathbf{p}) \neq 0$ (the zero set of a non-zero polynomial is always lower-dimensional). We ask: what is the rank of a generic tensor?

Warm-Up: Symmetric Matrices

We first answer this for $d = 2$, by counting parameters in the eigendecomposition. Consider $S = \sum_{i=1}^r \lambda_i \mathbf{v}_i \mathbf{v}_i^\top \in S^2(\mathbb{R}^p)$ with the $\mathbf{v}_i$ orthonormal (an eigendecomposition truncated to r terms).

Lemma 4.2. *The set of $S \in S^2(\mathbb{R}^p)$ with $\text{rank}(S) \leq r$ has dimension $rp - \binom{r}{2}$, for $0 \leq r \leq p$.*

Proof. Parametrize by $(\lambda_1, \dots, \lambda_r, \mathbf{v}_1, \dots, \mathbf{v}_r)$ with $\mathbf{v}_1, \dots, \mathbf{v}_r$ orthonormal. Choosing $\mathbf{v}_1$ (a unit vector, up to sign) costs $p - 1$ parameters; choosing $\mathbf{v}_2$ orthogonal to $\mathbf{v}_1$ (a unit vector in a $(p - 1)$ -dimensional space) costs $p - 2$; in general $\mathbf{v}_k$ costs $p - k$. Together with r parameters for $\lambda_1, \dots, \lambda_r$, the total is

$$r + \sum_{k=1}^r (p - k) = r + rp - \binom{r+1}{2} = rp - \binom{r}{2}. \quad \square$$

At $r = p$: $\dim = p^2 - \binom{p}{2} = \binom{p+1}{2} = \dim(S^2(\mathbb{R}^p))$. The rank $\leq p$ locus is the whole space. At $r = p - 1$:

$$\dim = (p - 1)p - \binom{p-1}{2} = \binom{p+1}{2} - 1,$$

one less than $\dim(S^2(\mathbb{R}^p))$: the rank $\leq p - 1$ locus is a hypersurface (indeed, it is $\{S : \det S = 0\}$, the zero set of one polynomial). Hence, since we know a rank p decomposition exists from the eigendecomposition,

Corollary 4.3. *A generic $S \in S^2(\mathbb{R}^p)$ has rank p .*

An alternate argument is to use the generic uniqueness of the eigendecomposition to conclude that a generic symmetric matrix cannot have an eigendecomposition in which one eigenvalue vanishes.

Symmetric Tensors of General Order

For $d \geq 3$ we redo the parameter count, this time for the unconstrained (not necessarily orthogonal) symmetric CP decomposition $T = \sum_{i=1}^r \lambda_i \mathbf{v}_i^{\otimes d}$ used to define symmetric rank.

Lemma 4.4. $\dim S^d(\mathbb{C}^n) = \binom{n+d-1}{d}$.

Proof. A symmetric tensor's entries are determined by, and freely chosen for, each multiset of d indices from $\{1, \dots, n\}$ (since permuting indices does not change the entry). Order such a multiset monotonically, $1 \leq i_1 \leq \dots \leq i_d \leq n$. Represent it as a string of d marks (one per index) and $n-1$ dividers, placed among $d+n-1$ positions in total, where the divider before position k separates values $< k$ from values $\geq k$: this is a bijection between monotone index strings and ways to choose which d of the $d+n-1$ positions are marks, giving $\binom{n+d-1}{d}$ monotone strings. $\square$

Writing λ_i 's scale into $\mathbf{v}_i$, a symmetric CP expression $T = \sum_{i=1}^r \mathbf{v}_i^{\otimes d}$ has rn free parameters (no orthogonality constraint imposed). Setting this equal to $\dim S^d(\mathbb{C}^n)$ suggests an *expected* generic rank of

$$r_{\text{exp}} := \left\lceil \frac{1}{n} \binom{n+d-1}{d} \right\rceil.$$

Remark 4.5 (Why this count fails for $d = 2$). Compare Lemma 4.4's count to the matrix warm-up: at $d = 2$, $r_{\text{exp}} = \lceil \binom{p+1}{2}/p \rceil \approx (p+1)/2$, less than the true generic rank p (Corollary 4.3). Given a rank- r symmetric matrix $M = WW^T$ ($W \in \mathbb{R}^{p \times r}$, $r \leq p$), every other factorization $M = W'(W')^T$ satisfies $W' = WQ$ for some Q in the orthogonal group $O(r)$ (the same computation as the ICA non-identifiability argument, Proposition 6.2 below): the unconstrained parametrization $(\mathbf{v}_1, \dots, \mathbf{v}_r) \mapsto \sum \mathbf{v}_i \mathbf{v}_i^T$ has, for every matrix in its image, a fiber of dimension $\dim O(r) = \binom{r}{2} > 0$. The naive count rn implicitly assumes a *finite* generic fiber, which fails here. For $d \geq 3$, by contrast, the generic fiber is finite (often a single point) for r below a threshold, so the naive count r_{exp} above is accurate for symmetric tensors of order ≥ 3 . Generic tensor decompositions, unlike generic matrix decompositions, are unique.

Theorem 4.6 (Alexander–Hirschowitz, 1995). *The symmetric rank of a generic $T \in S^d(\mathbb{C}^n)$ is*

$$\left\lceil \frac{1}{n} \binom{n+d-1}{d} \right\rceil,$$

except when $d = 2$ (where the generic rank is n , not the formula above), or in the special cases $(d, n) \in \{(3, 5), (4, 3), (4, 4), (4, 5)\}$, in which cases the generic rank is one more than the formula.

Theorem 4.6 was first conjectured in 1901; the case $d = 3$ proved in 1916, the general case by Alexander and Hirschowitz in 1995. See the reference for the complete argument.

Remark 4.7 (Open: the smallest unknown case for general secant dimensions). Theorem 4.6 settles the generic *symmetric* rank for every format. The analogous question for partially symmetric and unbalanced tensors, equivalently, the dimension (or defectivity) of secant varieties of Segre–Veronese varieties, Section 8, is not fully settled; H. Abo's work (with Ottaviani, Peterson, Brambilla, and others) resolves large families of cases by induction, but open cases remain.

Non-Symmetric Tensors

For a general (non-symmetric) CP expression $T = \sum_{i=1}^r \mathbf{a}_i^{(1)} \otimes \dots \otimes \mathbf{a}_i^{(d)}$, $\mathbf{a}_i^{(k)} \in \mathbb{C}^{n_k}$, the analogous count is $r(\sum_k n_k - (d-1))$ parameters (each term has n_k free entries per factor, minus $d-1$ redundant overall scalings), against $\dim \mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_d} = \prod_k n_k$. The expected generic rank $r_{\text{exp}} = \lceil \prod_k n_k / (\sum_k n_k - (d-1)) \rceil$ is not known to hold in general; unlike Theorem 4.6, no analogous exceptional-case classification for non-symmetric tensors is complete.

5 Moments and Cumulants

Recall from Module 1 that a random vector $\mathbf{z} \in \mathbb{R}^q$ has mean $\mu = \mathbb{E}[\mathbf{z}] \in \mathbb{R}^q$ and covariance $\Sigma_{\mathbf{z}} = \mathbb{E}[(\mathbf{z} - \mu)^{\otimes 2}] \in S^2(\mathbb{R}^q)$. Both are special cases of a general construction.

Definition 5.1 (Moment tensor). The d -th moment tensor of $\mathbf{z}$ is $\mathcal{M}_d(\mathbf{z}) := \mathbb{E}[\mathbf{z}^{\otimes d}] \in S^d(\mathbb{R}^q)$, with entries $\mathcal{M}_d(\mathbf{z})_{i_1 \dots i_d} = \mathbb{E}[z_{i_1} \cdots z_{i_d}]$.

$\mathcal{M}_1(\mathbf{z}) = \mu$; $\mathcal{M}_2(\mathbf{z}) = \mathbb{E}[\mathbf{z}\mathbf{z}^\top]$, the (uncentered) second moment matrix, related to the covariance by $\Sigma_{\mathbf{z}} = \mathcal{M}_2(\mathbf{z}) - \mu\mu^\top$.

Proposition 5.2. If $z_1, \dots, z_q$ are independent with mean zero, $\mathcal{M}_3(\mathbf{z})$ is diagonal, i.e. $\mathcal{M}_3(\mathbf{z})_{ijk} = 0$ unless $i = j = k$.

Proof. Independence gives $\mathbb{E}[z_i z_j z_k] = \mathbb{E}[z_i]\mathbb{E}[z_j]\mathbb{E}[z_k]$ whenever i, j, k are not all equal. If i, j, k are pairwise distinct, this is $0 \cdot 0 \cdot 0 = 0$. If two coincide, say $j = k \neq i$, then $\mathbb{E}[z_i z_j^2] = \mathbb{E}[z_i]\mathbb{E}[z_j^2] = 0$ since $\mathbb{E}[z_i] = 0$. $\square$

Remark 5.3 (The same argument fails at $d = 4$). The entry $\mathcal{M}_4(\mathbf{z})_{iijj}$ ($i \neq j$) equals $\mathbb{E}[z_i^2 z_j^2] = \mathbb{E}[z_i^2]\mathbb{E}[z_j^2]$ by independence, which is generically *non-zero*, since it is a product of two non-negative quantities. So $\mathcal{M}_4(\mathbf{z})$ is not diagonal in general, even though $\mathcal{M}_3(\mathbf{z})$ is: independence of the coordinates of $\mathbf{z}$ does not, on its own, diagonalize every moment tensor. This is the reason we use *cumulants*, developed below, rather than moments.

Moments transform under a linear map by a direct generalization of Module 1's congruence action $\Sigma \mapsto A\Sigma A^\top$.

Definition 5.4 (The action of A on a symmetric tensor). For $A \in \mathbb{R}^{p \times q}$ and $S \in S^d(\mathbb{R}^q)$, define $A \bullet S \in S^d(\mathbb{R}^p)$ by

$$(A \bullet S)_{i_1 \dots i_d} = \sum_{\alpha_1, \dots, \alpha_d=1}^q S_{\alpha_1 \dots \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d}.$$

At $d = 2$, $(A \bullet S)_{ij} = \sum_{\alpha\beta} S_{\alpha\beta} A_{i\alpha} A_{j\beta} = (ASA^\top)_{ij}$: this recovers the congruence action.

Proposition 5.5 (Multilinearity). Let $\mathbf{x} = A\mathbf{z}$ for $A \in \mathbb{R}^{p \times q}$. Then $\mathcal{M}_d(\mathbf{x}) = A \bullet \mathcal{M}_d(\mathbf{z})$.

Proof.

$$\mathcal{M}_d(\mathbf{x})_{i_1 \dots i_d} = \mathbb{E}[x_{i_1} \cdots x_{i_d}] = \mathbb{E}\left[\prod_{k=1}^d \sum_{\alpha_k} A_{i_k \alpha_k} z_{\alpha_k}\right] = \sum_{\alpha_1, \dots, \alpha_d} A_{i_1 \alpha_1} \cdots A_{i_d \alpha_d} \mathbb{E}[z_{\alpha_1} \cdots z_{\alpha_d}],$$

using linearity of expectation to move it past the (finite) sum over $\alpha_1, \dots, \alpha_d$; the last factor is $\mathcal{M}_d(\mathbf{z})_{\alpha_1 \dots \alpha_d}$, giving $(A \bullet \mathcal{M}_d(\mathbf{z}))_{i_1 \dots i_d}$. $\square$

Generating Functions

Definition 5.6 (Moment and cumulant generating functions). For $\mathbf{t} = (t_1, \dots, t_q)^\top$, the *moment generating function* (mgf) of $\mathbf{z}$ is $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp\langle \mathbf{t}, \mathbf{z} \rangle]$, and the *cumulant generating function* (cgf) is $K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t})$.

Expanding $\exp\langle \mathbf{t}, \mathbf{z} \rangle = \sum_{d \geq 0} \frac{1}{d!} \langle \mathbf{t}, \mathbf{z} \rangle^d$ and using $\mathbb{E}[\langle \mathbf{t}, \mathbf{z} \rangle^d] = \langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z}) \rangle$ (expand $\langle \mathbf{t}, \mathbf{z} \rangle^d = \sum_{i_1, \dots, i_d} t_{i_1} \cdots t_{i_d} z_{i_1} \cdots z_{i_d}$ and take expectations termwise) gives

$$M_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 0} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{M}_d(\mathbf{z}) \rangle.$$

Define the d -th cumulant tensor $\mathcal{K}_d(\mathbf{z}) \in S^d(\mathbb{R}^q)$ as the analogous coefficient of $K_{\mathbf{z}}$:

$$K_{\mathbf{z}}(\mathbf{t}) = \sum_{d \geq 1} \frac{1}{d!} \langle \mathbf{t}^{\otimes d}, \mathcal{K}_d(\mathbf{z}) \rangle.$$

Proposition 5.7. $\mathcal{K}_1(\mathbf{z}) = \mu$ and $\mathcal{K}_2(\mathbf{z}) = \Sigma_{\mathbf{z}}$.

Proof. Write $K_{\mathbf{z}}(\mathbf{t}) = \log(1 + u)$ with $u = \langle \mathbf{t}, \mu \rangle + \frac{1}{2} \langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z}) \rangle + O(\mathbf{t}^3)$, and expand $\log(1 + u) = u - \frac{1}{2}u^2 + O(u^3)$. The degree-1 part of $\log(1 + u)$ is $\langle \mathbf{t}, \mu \rangle$, giving $\mathcal{K}_1(\mathbf{z}) = \mu$. The degree-2 part is $\frac{1}{2} \langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z}) \rangle - \frac{1}{2} \langle \mathbf{t}, \mu \rangle^2 = \frac{1}{2} \langle \mathbf{t}^{\otimes 2}, \mathcal{M}_2(\mathbf{z}) - \mu^{\otimes 2} \rangle$ (using $\langle \mathbf{t}, \mu \rangle^2 = \langle \mathbf{t}^{\otimes 2}, \mu^{\otimes 2} \rangle$), so $\mathcal{K}_2(\mathbf{z}) = \mathcal{M}_2(\mathbf{z}) - \mu^{\otimes 2} = \mathbb{E}[(\mathbf{z} - \mu)^{\otimes 2}] = \Sigma_{\mathbf{z}}$. $\square$

Proposition 5.8 (Cumulants diagonalize under independence, at every order). *If $z_1, \dots, z_q$ are independent, $\mathcal{K}_d(\mathbf{z})$ is diagonal for every d .*

Proof. Independence gives $M_{\mathbf{z}}(\mathbf{t}) = \mathbb{E}[\exp(t_1 z_1) \cdots \exp(t_q z_q)] = \prod_{i=1}^q \mathbb{E}[\exp(t_i z_i)]$, so

$$K_{\mathbf{z}}(\mathbf{t}) = \log M_{\mathbf{z}}(\mathbf{t}) = \sum_{i=1}^q \log \mathbb{E}[\exp(t_i z_i)],$$

a sum of functions each depending on a *single* coordinate t_i . Consequently no monomial $t_i t_j \cdots$ with two or more distinct indices can appear in the Taylor expansion of $K_{\mathbf{z}}$ at any degree, i.e. the degree- d coefficient tensor $\mathcal{K}_d(\mathbf{z})$ has entries supported only on indices $(i, i, \dots, i)$. $\square$

Proposition 5.8 is the resolution of Remark 5.3: cumulants, unlike moments, diagonalize under independence at *every* order, not just order 3. Passing from moments to cumulants can be thought of as the higher-order analog of mean-centering.

6 Independent Component Analysis

Definition 6.1 (Independent Component Analysis). *Independent Component Analysis* (ICA) models an observed random vector $\mathbf{x} = (x_1, \dots, x_p)^{\top}$ as $\mathbf{x} = A\mathbf{z}$ for a fixed, unknown, invertible mixing matrix $A \in \mathbb{R}^{p \times p}$ and a latent random vector $\mathbf{z} = (z_1, \dots, z_p)^{\top}$ with *independent* entries. (We take $p = q$, the classical case; Section 8 discusses $q \neq p$.)

This is Module 1's linear latent-variable model $\mathbf{x} = A\mathbf{z}$, with the entries of $\mathbf{z}$ independent rather than just uncorrelated. The classical motivating picture is the *cocktail party problem*: p microphones each record a mixture $x_i = \sum_j A_{ij} z_j$ of p underlying, independent sound sources z_j ; the goal is to recover A and the sources from the microphone recordings alone (blind source separation).

Why the Covariance Alone Is Not Enough

Since the z_j are independent, $\Sigma_{\mathbf{z}}$ is diagonal, so $\Sigma_{\mathbf{x}} = A\Sigma_{\mathbf{z}}A^\top = ADA^\top$ for some diagonal $D \succeq 0$. Given (an estimate of) $\Sigma_{\mathbf{x}}$, can we recover A and D ?

Proposition 6.2. *Let $\Sigma \in S^2(\mathbb{R}^p)$, $\Sigma \succeq 0$. The solution set $\{(A, D) : A \in \mathbb{R}^{p \times p} \text{ invertible}, D \succeq 0 \text{ diagonal}, \Sigma = ADA^\top\}$, if non-empty, is*

$$\{(B_0 Q \Lambda, \Lambda^{-2}) : Q \in O(p), \Lambda \text{ diagonal invertible}\}$$

for any one solution $B_0 = A_0 D_0^{1/2}$.

Proof. Write $B = AD^{1/2}$, so $\Sigma = BB^\top$. If $B_0 B_0^\top = BB^\top = \Sigma$, then $(B_0^{-1}B)(B_0^{-1}B)^\top = B_0^{-1}\Sigma(B_0^\top)^{-1} = B_0^{-1}(B_0 B_0^\top)(B_0^\top)^{-1} = I$, so $Q := B_0^{-1}B$ is orthogonal and $B = B_0 Q$. Since $B = AD^{1/2}$, any positive rescaling of B 's columns can be absorbed into $D^{1/2}$; writing this rescaling as Λ gives $(A, D) = (B_0 Q \Lambda, \Lambda^{-2})$. $\square$

The solution set has dimension $\dim O(p) + p = \binom{p}{2} + p$ (an arbitrary rotation, plus rescaling each of p columns), strictly positive for $p \geq 2$: $\Sigma_{\mathbf{x}}$ alone never pins down (A, D) . As a consistency check: the naive parametrization (A, D) has $p^2 + p$ free entries, mapping into $\Sigma \in S^2(\mathbb{R}^p)$, of dimension $\binom{p+1}{2}$; a generic fiber of a map between spaces of these dimensions is expected to have dimension $(p^2 + p) - \binom{p+1}{2} = \binom{p}{2} + p$, matching the solution set above.

Diagonalizing a Higher Cumulant Instead

Recovering A requires looking beyond $\Sigma_{\mathbf{x}}$. Perhaps, then, we can ammend Benzécri's statement to say that “all data analysis is tensor diagonalization; the crux of the matter is in finding the right tensor to diagonalize.” Having diagonalized the covariance, we now diagonalize a higher cumulant.

Proposition 6.3. *Fix $d \geq 3$. Writing $A = [\mathbf{a}_1 \cdots \mathbf{a}_p]$ (columns) and $\lambda_i := \mathcal{K}_d(\mathbf{z})_{i,i,\dots,i}$ (d copies of i ; the i -th source's own order- d cumulant),*

$$\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^p \lambda_i \mathbf{a}_i^{\otimes d}.$$

Proof. By Proposition 5.8, $\mathcal{K}_d(\mathbf{z}) = \sum_{i=1}^p \lambda_i \mathbf{e}_i^{\otimes d}$, diagonal. By the cumulant analog of multilinearity of moments (Proposition 5.5; the identical proof applies to cgf Taylor coefficients), $\mathcal{K}_d(\mathbf{x}) = A \bullet \mathcal{K}_d(\mathbf{z}) = \sum_i \lambda_i (A \bullet \mathbf{e}_i^{\otimes d})$. Directly from the definition of $\bullet$, $(A \bullet \mathbf{e}_i^{\otimes d})_{j_1 \dots j_d} = A_{j_1 i} \cdots A_{j_d i} = (\mathbf{a}_i^{\otimes d})_{j_1 \dots j_d}$, so $A \bullet \mathbf{e}_i^{\otimes d} = \mathbf{a}_i^{\otimes d}$. $\square$

Proposition 6.3 is the point of this module: *ICA is a symmetric tensor decomposition problem*. Recovering A from $\mathbf{x}$'s d -th cumulant means recovering the rank-one terms of a symmetric CP decomposition, and the columns of A are its factors. Sections 7 and 8 ask when the decomposition is unique.

7 Identifiability of ICA

The best we can hope to recover is A up to permuting and rescaling its columns: if $\mathbf{x} = A\mathbf{z}$, then for any permutation matrix P and invertible diagonal Λ , $\mathbf{x} = (AP\Lambda)(\Lambda^{-1}P^\top \mathbf{z})$, and $\Lambda^{-1}P^\top \mathbf{z}$ is again a vector of independent variables. So “identifiable” means: recoverable up to this ambiguity.

Definition 7.1. A random variable z is *Gaussian* with mean μ , variance σ^2 if it has density $f(x) = (2\pi\sigma^2)^{-1/2} \exp(-\frac{1}{2\sigma^2}(x - \mu)^2)$; write $z \sim N(\mu, \sigma^2)$.

Theorem 7.2 (Marcinkiewicz, 1939). *A Gaussian variable $z \sim N(\mu, \sigma^2)$ has cgf $K_z(t) = \mu t + \frac{1}{2}\sigma^2 t^2$. Conversely, if a random variable has a cumulant generating function that is finite for all t and is a polynomial, then it is Gaussian.*

A Gaussian variable has $\kappa_d(z) = 0$ for every $d \geq 3$. By Proposition 6.3, if every source z_i is Gaussian, $\mathcal{K}_d(\mathbf{x}) = 0$ for every $d \geq 3$: Gaussian sources are invisible to the higher-order statistics. This is an obstruction to identifiability.

Setting Up Non-Identifiability

Suppose $\mathbf{x} = A\mathbf{z} = A'\mathbf{z}'$ for two invertible mixing matrices and two vectors of independent sources. Then $\mathbf{z} = (A^{-1}A')\mathbf{z}' =: B\mathbf{z}'$, so B is an invertible matrix sending one vector of independent variables, $\mathbf{z}'$, to another, $\mathbf{z}$.

Theorem 7.3 (Darmois–Skitovich, 1953). *Let $\mathbf{z} = (z_1, \dots, z_p)$ be independent, and let $\sum_i a_i z_i$ and $\sum_i b_i z_i$ be independent (for fixed scalars a_i, b_i). Then every z_i with $a_i b_i \neq 0$ is Gaussian.*

Proposition 7.4. *Let $\mathbf{z}' = (z'_1, \dots, z'_p)$ be independent and $B \in \mathbb{R}^{p \times p}$ invertible, with $B\mathbf{z}'$ also independent. Then for every source index i , either z'_i is Gaussian, or column i of B has exactly one non-zero entry.*

Proof. For any two rows $j \neq k$, $(B\mathbf{z}')_j = \sum_i B_{ji} z'_i$ and $(B\mathbf{z}')_k = \sum_i B_{ki} z'_i$ are independent (two coordinates of an independent vector). By Theorem 7.3, for every i : either $B_{ji} B_{ki} = 0$, or z'_i is Gaussian. If z'_i is not Gaussian, this holds for *every* pair $j \neq k$, which forces at most one row j with $B_{ji} \neq 0$ (two non-zero rows j, k in column i would violate $B_{ji} B_{ki} = 0$). Since B is invertible, column i is non-zero, so it has exactly one non-zero entry. $\square$

Theorem 7.5 (Comon, 1994). *Fix $\mathbf{x} = A\mathbf{z}$, A invertible, with $\mathbf{z} = (z_1, \dots, z_p)$ independent. The ICA model is identifiable (i.e. A is recoverable up to permutation and scaling of columns) if and only if at most one z_i is Gaussian.*

We prove Theorem 7.5 in three parts, following the setup above: $B = A^{-1}A'$ is invertible and sends the independent vector $\mathbf{z}'$ to the independent vector $\mathbf{z} = B\mathbf{z}'$.

Part 1: no Gaussian sources $\Rightarrow$ identifiable. By Proposition 7.4, every column of B has one non-zero entry, say column i 's non-zero entry is in row $r(i)$. If $r(i) = r(i')$ for $i \neq i'$, columns i, i' are both scalar multiples of $\mathbf{e}_{r(i)}$, hence linearly dependent, contradicting invertibility of B ; so r is injective, hence (as a self-map of a finite set) a bijection. So $B = P\Lambda$ for a permutation matrix P and invertible diagonal Λ : the trivial ambiguity. $\square$

Part 2: two or more Gaussian sources $\Rightarrow$ not identifiable. Suppose z'_1, z'_2 are both Gaussian and independent; rescale so both are $N(0, 1)$. The linear map $B_0 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ (identity on all other coordinates) sends (z'_1, z'_2) to $(z'_1 + z'_2, z'_1 - z'_2)$, jointly Gaussian (a linear map of a Gaussian vector) with covariance $B_0 B_0^\top = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. A jointly Gaussian vector with diagonal covariance has independent entries: its density $f(\mathbf{x}) \propto \exp(-\frac{1}{2}\mathbf{x}^\top \Sigma^{-1} \mathbf{x})$ factors into a product of one-dimensional Gaussian densities when Σ (equivalently Σ^{-1}) is diagonal. So $B_0 \mathbf{z}'$ is again a vector of independent variables, but B_0 is not a signed permutation: this is a different, valid solution, so A is not identifiable. (If $p > 2$, extend B_0 by the identity on the remaining coordinates; the argument is unchanged.) $\square$

Part 3: one Gaussian source $\Rightarrow$ identifiable. Reorder so z'_p is the (unique) Gaussian source; $z'_1, \dots, z'_{p-1}$ are not Gaussian. By Proposition 7.4, columns $1, \dots, p-1$ of B each have one non-zero entry, and (as in Part 1) these must lie in $p-1$ distinct rows; permute rows so column i 's non-zero entry is in row i , for $i = 1, \dots, p-1$, and rescale rows so each such entry equals 1. Since B is invertible and columns $1, \dots, p-1$ already occupy rows $1, \dots, p-1$ (each with a single entry), row p must be zero in columns $1, \dots, p-1$, and column p has an entry $\lambda_p \neq 0$ in row p (else B singular). Write column p 's other entries as $\lambda_1, \dots, \lambda_{p-1}$, so

$$(B\mathbf{z}')_i = z'_i + \lambda_i z'_p \quad (i < p), \quad (B\mathbf{z}')_p = \lambda_p z'_p.$$

Since $B\mathbf{z}'$ is independent, $(B\mathbf{z}')_i \perp (B\mathbf{z}')_p$ for each $i < p$, in particular uncorrelated:

$$0 = \text{Cov}(z'_i + \lambda_i z'_p, \lambda_p z'_p) = \lambda_p [\text{Cov}(z'_i, z'_p) + \lambda_i \text{Var}(z'_p)] = \lambda_p \lambda_i \text{Var}(z'_p),$$

using $z'_i \perp z'_p$. Since $\lambda_p \neq 0$ and $\text{Var}(z'_p) > 0$ (non-degenerate), $\lambda_i = 0$ for every $i < p$. So B (after undoing the row permutation) is diagonal, hence a signed permutation. $\square$

8 Existence and Uniqueness of Tensor Decompositions

In practice we have only a data matrix $X \in \mathbb{R}^{n \times p}$, and must estimate $\mathcal{K}_d(\mathbf{x})$ from samples before decomposing it. This raises a different question: does the tensor $\mathcal{K}_d(\mathbf{x}) = \sum_{i=1}^p \lambda_i \mathbf{a}_i^{\otimes d}$, viewed simply as a symmetric tensor in $S^d(\mathbb{R}^p)$ of rank p , have a *unique* such decomposition?

The number of equations available (matching entries of $\mathcal{K}_d(\mathbf{x})$) is $\dim S^d(\mathbb{R}^p) = \binom{p+d-1}{d}$; the number of unknowns in A (with $q \leq p$ sources) is qp . Larger d gives more equations relative to unknowns, at the cost of a d -th cumulant that is harder to estimate accurately from finite samples. This is a practical tradeoff in choosing d .

Theorem 8.1 (Harshman, 1970). *Let $T = \sum_{i=1}^r \lambda_i \mathbf{a}_i^{\otimes d} \in S^d(\mathbb{R}^p)$, $A = [\mathbf{a}_1 \cdots \mathbf{a}_r]$. If $r = \text{rank}(A)$ (in particular $r \leq p$), this decomposition is the unique one, up to permutation and rescaling of terms.*

Exercise 8.2. Let $T = \mathbf{e}_1^{\otimes 3} + \mathbf{e}_2^{\otimes 3} \in S^3(\mathbb{R}^p)$, $p \geq 2$. Identify the factor matrix A to which Theorem 8.1 applies, and check $\text{rank}(A) = 2$.

Theorem 8.1's hypothesis $r = \text{rank}(A)$ is restrictive: it requires A 's columns to be linearly independent, hence $r \leq p$. A weaker, more robust condition suffices.

Definition 8.3 (Kruskal rank). The *Kruskal rank* of $A \in \mathbb{R}^{p \times r}$, denoted k_A , is the largest integer k such that every k columns of A are linearly independent.

Always $k_A \leq \text{rank}(A) \leq p$; unlike ordinary rank, k_A can be strictly less than $\text{rank}(A)$ even for a full-rank A with $r > p$ columns (e.g. four columns in $\mathbb{R}^3$, no three of which are dependent, but some four necessarily are).

Theorem 8.4 (Kruskal, 1977). *If $r \leq \frac{d}{2}k_A - 1$, the decomposition $T = \sum_{i=1}^r \lambda_i \mathbf{a}_i^{\otimes d}$ is the unique one, up to permutation and rescaling of terms.*

We cite Theorem 8.4 without proof; it is a useful general-purpose sufficient condition for uniqueness of a tensor decomposition.

Generic Uniqueness

Theorems 8.1 and 8.4 certify uniqueness for a *specific* tensor, given information about its factor matrix. A different, and in this module’s context more striking, question is whether uniqueness holds for *almost every* tensor of a given rank, with no assumption on the factors at all.

Theorem 8.5 (Chiantini–Ottaviani–Vannieuwenhoven, 2017). *A generic $T \in S^d(\mathbb{C}^n)$ of rank strictly less than the Alexander–Hirschowitz threshold $\frac{1}{n}\binom{n+d-1}{d}$ (Theorem 4.6) has a unique symmetric rank decomposition, except for three specific formats, $S^6(\mathbb{C}^3)$ at rank 9, $S^4(\mathbb{C}^4)$ at rank 8, $S^3(\mathbb{C}^6)$ at rank 9, each of which has two decompositions.*

Together with Remark 4.5, this is the module’s central identifiability contrast with Module 1: a *generic* low-rank symmetric tensor of order $d \geq 3$ is already uniquely decomposable, with no extra structural assumption (such as orthogonality) required, unlike a symmetric matrix, whose generic decomposition always has a positive-dimensional family of alternatives (Remark 4.5).

The Algebraic Geometry of Secant Varieties

Theorem 8.5 is a statement in classical algebraic geometry. We record the minimal vocabulary needed to state it precisely.

Definition 8.6. A *variety* is a subset of $\mathbb{C}^N$ defined by the vanishing of a collection of polynomial equations. A *limit point* of a set X is a point of the form $\lim_{\varepsilon \rightarrow 0} \mathbf{x}_\varepsilon$ for a convergent sequence $\mathbf{x}_\varepsilon \in X$; the *closure* $\overline{X}$ of X is X together with all its limit points.

Definition 8.7. The *Veronese variety* $\nu_d(\mathbb{C}^n) \subseteq S^d(\mathbb{C}^n)$ is the set of symmetric tensors of rank ≤ 1 . The *Segre variety* $\text{Seg}(\mathbb{C}^{n_1} \times \cdots \times \mathbb{C}^{n_d}) \subseteq \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d}$ is the set of (not necessarily symmetric) tensors of rank ≤ 1 .

Definition 8.8. The *r -th secant variety* of the Veronese, $\sigma_r(\nu_d(\mathbb{C}^n)) \subseteq S^d(\mathbb{C}^n)$, is the closure of the set of symmetric tensors of rank $\leq r$. Likewise, $\sigma_r(\text{Seg}(\mathbb{C}^{n_1} \times \cdots \times \mathbb{C}^{n_d}))$ is the closure of the set of tensors of rank $\leq r$.

Why is the closure needed: why can a tensor’s rank jump under a limit? The standard example: for $\mathbf{v} = \mathbf{e}_1, \mathbf{w} = \mathbf{e}_2 \in \mathbb{C}^2$,

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left[(\mathbf{v} + \varepsilon \mathbf{w})^{\otimes 3} - \mathbf{v}^{\otimes 3} \right] = \mathbf{v} \otimes \mathbf{v} \otimes \mathbf{w} + \mathbf{v} \otimes \mathbf{w} \otimes \mathbf{v} + \mathbf{w} \otimes \mathbf{v} \otimes \mathbf{v},$$

by expanding $(\mathbf{v} + \varepsilon \mathbf{w})^{\otimes 3} = \mathbf{v}^{\otimes 3} + \varepsilon(\mathbf{v} \otimes \mathbf{v} \otimes \mathbf{w} + \mathbf{v} \otimes \mathbf{w} \otimes \mathbf{v} + \mathbf{w} \otimes \mathbf{v} \otimes \mathbf{v}) + O(\varepsilon^2)$. For each $\varepsilon \neq 0$, the left-hand bracket, before dividing by ε , is a difference of two rank-one tensors, hence of rank ≤ 2 ; the limiting tensor $T = \mathbf{v} \otimes \mathbf{v} \otimes \mathbf{w} + \mathbf{v} \otimes \mathbf{w} \otimes \mathbf{v} + \mathbf{w} \otimes \mathbf{v} \otimes \mathbf{v}$ therefore lies in $\sigma_2(\text{Seg}(\mathbb{C}^2 \times \mathbb{C}^2 \times \mathbb{C}^2))$, yet it is a classical fact that $\text{rank}(T) = 3$: T itself is *not* a sum of 2 rank-one tensors, only a *limit* of such sums. This is the phenomenon behind a problem on Homework 2: a rank-3 tensor exhibited as a limit of rank-2 tensors. Without the closure, σ_2 would miss T entirely, even though T is, in every meaningful sense, “approximately rank 2.”

9 Optimization Landscapes for Tensors

Existence (Theorem 4.6) and uniqueness (Theorems 8.1–8.5) leave the module’s third question unanswered: how do you actually *find* a tensor decomposition? Recall from Module 1 that the top principal component is the maximizer of $\mathbf{u}^\top \mathbf{S} \mathbf{u}$ over $\|\mathbf{u}\| = 1$, and that its Lagrangian stationarity condition, $\mathbf{S} \mathbf{u} = \lambda \mathbf{u}$, is the eigenvector equation. We generalize this to symmetric tensors.

Proposition 9.1 (Critical points of $T(\mathbf{v}, \mathbf{v}, \mathbf{v})$). *Let $T \in S^3(\mathbb{R}^n)$ and consider $\max_{\|\mathbf{v}\|=1} T(\mathbf{v}, \mathbf{v}, \mathbf{v})$, where $T(\mathbf{v}, \mathbf{v}, \mathbf{v}) = \sum_{ijk} T_{ijk} v_i v_j v_k$. The critical points of the Lagrangian are the $\mathbf{v}$ satisfying*

$$T(\cdot, \mathbf{v}, \mathbf{v}) \parallel \mathbf{v}, \quad \text{i.e.} \quad \sum_{jk} T_{ljk} v_j v_k = \mu v_l \quad \forall l,$$

for some scalar μ .

Proof. The Lagrangian is $L(\mathbf{v}, \mu) = \sum_{ijk} T_{ijk} v_i v_j v_k - \mu(\|\mathbf{v}\|^2 - 1)$. Differentiating the cubic form by the product rule,

$$\frac{\partial}{\partial v_l} \sum_{ijk} T_{ijk} v_i v_j v_k = \sum_{jk} T_{ljk} v_j v_k + \sum_{ik} T_{ilk} v_i v_k + \sum_{ij} T_{ijl} v_i v_j = 3 \sum_{jk} T_{ljk} v_j v_k,$$

where the three sums are equal by symmetry of T (relabel dummy indices in the second and third sums). So $\partial L / \partial v_l = 3 \sum_{jk} T_{ljk} v_j v_k - 2\mu v_l = 0$ for all l , i.e. $T(\cdot, \mathbf{v}, \mathbf{v}) = \frac{2\mu}{3} \mathbf{v}$: proportional to $\mathbf{v}$. $\square$

Definition 9.2 (Eigenvector of a tensor). A unit vector $\mathbf{v}$ with $T(\cdot, \mathbf{v}, \mathbf{v}) = \mu \mathbf{v}$ for some scalar μ is an *eigenvector* of $T \in S^3(\mathbb{R}^n)$, with eigenvalue μ . (The same definition, with the identical proof, applies to $T \in S^d(\mathbb{R}^n)$ for any $d \geq 2$; at $d = 2$ it recovers the matrix eigenvector equation $S\mathbf{v} = \mu \mathbf{v}$.)

Proposition 9.3 (Eigenvectors give the best rank-one approximation). *The unit vector $\mathbf{v}$ maximizing $T(\mathbf{v}, \mathbf{v}, \mathbf{v})^2$ also minimizes $\|T - \lambda \mathbf{v}^{\otimes 3}\|$ over $\|\mathbf{v}\| = 1$, $\lambda \in \mathbb{R}$; both are eigenvectors of T .*

Proof. For unit $\mathbf{v}$, $\|T - \lambda \mathbf{v}^{\otimes 3}\|^2 = \langle T, T \rangle - 2\lambda \langle T, \mathbf{v}^{\otimes 3} \rangle + \lambda^2 \langle \mathbf{v}^{\otimes 3}, \mathbf{v}^{\otimes 3} \rangle = \|T\|^2 - 2\lambda T(\mathbf{v}, \mathbf{v}, \mathbf{v}) + \lambda^2$, using $\|\mathbf{v}\| = 1$. Minimizing over λ : $\partial_\lambda(\dots) = -2T(\mathbf{v}, \mathbf{v}, \mathbf{v}) + 2\lambda = 0 \Rightarrow \lambda = T(\mathbf{v}, \mathbf{v}, \mathbf{v})$, and substituting back leaves $-T(\mathbf{v}, \mathbf{v}, \mathbf{v})^2$ to minimize, i.e. $T(\mathbf{v}, \mathbf{v}, \mathbf{v})^2$ to maximize. Since $T(\mathbf{v}, \mathbf{v}, \mathbf{v})^2$ is a smooth non-negative function of $\mathbf{v}$ whose critical points coincide with those of $T(\mathbf{v}, \mathbf{v}, \mathbf{v})$ itself (by the chain rule, away from $T(\mathbf{v}, \mathbf{v}, \mathbf{v}) = 0$), its maximizer is an eigenvector of T by Proposition 9.1. $\square$

Remark 9.4 (A novelty of odd order). Since 3 is odd, $T(-\mathbf{v}, -\mathbf{v}, -\mathbf{v}) = -T(\mathbf{v}, \mathbf{v}, \mathbf{v})$. If $T(\mathbf{v}, \mathbf{v}, \mathbf{v}) \geq 0$ held for *every* unit $\mathbf{v}$, applying this to $-\mathbf{v}$ would force $T(\mathbf{v}, \mathbf{v}, \mathbf{v}) \leq 0$ too, hence $T(\mathbf{v}, \mathbf{v}, \mathbf{v}) = 0$ for every $\mathbf{v}$, which, by Proposition 3.1, forces $T = 0$. So a non-zero $T \in S^3(\mathbb{R}^n)$ can *never* be made sign-definite, no matter its structure. This is why Proposition 9.3 maximizes $T(\mathbf{v}, \mathbf{v}, \mathbf{v})^2$, not $T(\mathbf{v}, \mathbf{v}, \mathbf{v})$ itself, in contrast to the matrix case ($d = 2$), where $(-\mathbf{v})^\top S(-\mathbf{v}) = \mathbf{v}^\top S \mathbf{v}$ (no sign flip), and a PSD matrix achieves $\mathbf{v}^\top S \mathbf{v} \geq 0$ for every $\mathbf{v}$ (Module 1).

Remark 9.5 (Singular vectors of general tensors). The same variational idea, without symmetry, defines *singular vectors* of a general $T \in \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2} \otimes \mathbb{R}^{n_3}$: unit vectors $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ that are critical points of $T(\mathbf{u}, \mathbf{v}, \mathbf{w})$ subject to $\|\mathbf{u}\| = \|\mathbf{v}\| = \|\mathbf{w}\| = 1$, directly generalizing the singular vector pairs of Module 1's SVD. We do not derive this further here.

10 Outlook: Beyond Tensor Decomposition

Module 1 closed by asking what other matrices are worth diagonalizing, and named the fourth cumulant, a $p \times p \times p \times p$ array, as a first example. This module has extended this: example. We have studied the multi-linear algebra of tensors, developed from three definitions of a tensor, gives the vocabulary needed to diagonalize them, and studied ICA as the resulting model. Its identifiability rests on a new phenomenon, absent from the matrix case: a generic low-rank tensor

of order ≥ 3 is already uniquely decomposable (Theorem 8.5), where a generic symmetric matrix never is (Remark 4.5).

This phenomenon is not specific to ICA. Spectral and moment-based methods for estimating latent-variable and mixture models use it: build a higher moment or cumulant tensor of the observed data, and decompose it, using Kruskal- or genericity-type uniqueness (Theorems 8.4, 8.5) to recover parameters that pairwise, covariance-level statistics cannot.

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